

DEDOUGOU, Burkina Faso

WMO: 655050

Lat: 12.467N

Lon: 3.483W

Elev: 984

StdP: 14.18

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	63.4	65.6	15.4	12.4	85.7	19.0	14.8	85.3	16.6	81.7	15.3	81.7	3.5	40	0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	20.1	105.2	69.0	103.7	68.9	102.3	69.0	80.0	89.6	79.2	89.3	78.6	89.0	6.3	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.8	149.7	84.4	76.9	145.1	83.7	76.1	141.7	83.2	44.4	89.5	43.6	89.3	42.9	88.7	89.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	DB	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.6	14.3	12.5 <td>WB</td> <td>N/A</td> <td>N/A</td> <td>N/A</td> <td>N/A</td> <td>N/A</td> <td>N/A</td> <td>N/A</td> <td>N/A</td> <td>N/A</td> <td>N/A</td> <td>N/A</td> <td>N/A</td>		WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
				45.3	82.0	1.4	0.8	44.2	82.6	43.4	83.1	42.6	83.6	41.6	84.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	84.3	78.3	83.6	89.7	91.9	90.6	86.0	81.7	79.8	81.1	85.1	84.4	79.3
	DBStd	5.49	3.45	4.57	3.67	3.52	3.87	3.93	3.09	2.33	2.54	2.27	2.20	3.27
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12508	878	940	1230	1258	1257	1081	982	922	933	1087	1032	908
	CDD65	7033	413	520	765	808	792	631	517	457	483	622	582	443
	CDH74	93098	4746	7034	12110	13008	12145	8594	5604	4144	4889	7772	7849	5203
	CDH80	52597	2306	4054	7932	8759	7822	4722	2310	1318	1999	4154	4582	2639
(14) Wind	WSAvg	5.7	6.2	6.5	5.9	6.3	6.9	6.8	5.7	4.6	4.0	4.4	4.4	6.2
(15) Precipitation	PrecAvg	31.8	0.0	0.0	0.2	0.9	2.9	4.1	7.5	9.0	5.1	1.7	0.1	0.0
	PrecMax	45.3	0.0	1.2	2.0	7.4	7.9	8.7	13.1	19.4	9.6	6.5	0.7	0.3
	PrecMin	20.2	0.0	0.0	0.0	0.0	0.3	1.5	3.5	4.2	1.9	0.0	0.0	0.0
	PrecStd	6.6	0.0	0.2	0.5	1.4	1.8	1.6	2.6	3.2	2.1	1.5	0.2	0.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.1	103.3	106.3	107.6	106.7	102.2	96.3	91.0	94.7	100.0	100.5	97.9
		MCWB	61.8	64.5	66.0	69.2	71.6	76.3	77.2	78.9	78.1	70.3	67.0	63.0
	2%	DB	94.9	101.0	104.4	105.3	104.4	99.6	93.4	89.4	92.7	98.2	99.1	95.8
		MCWB	61.2	64.1	66.3	69.1	72.9	76.8	76.9	78.3	78.4	71.9	66.7	62.2
	5%	DB	92.5	99.0	103.0	103.9	102.4	97.2	91.2	87.9	90.9	96.6	97.9	93.8
		MCWB	60.0	63.2	66.1	69.1	73.6	76.8	76.8	77.8	78.3	73.6	66.4	61.6
	10%	DB	90.4	96.6	101.5	102.4	100.3	95.1	89.3	86.3	89.0	94.8	96.4	91.7
		MCWB	59.1	62.2	65.7	69.4	74.5	76.5	76.8	77.4	77.9	74.4	66.2	61.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.7	71.9	75.4	78.1	79.7	80.0	79.4	80.1	81.3	80.8	75.5	68.0
		MCDB	83.4	91.4	95.0	94.3	94.0	92.3	88.2	87.9	90.1	89.3	90.6	89.4
	2%	WB	63.4	68.5	74.0	76.8	78.3	79.1	78.7	79.2	80.1	79.3	73.8	64.9
		MCDB	90.7	91.2	93.1	94.1	93.8	91.9	88.6	86.9	88.9	89.3	88.4	91.1
	5%	WB	61.5	65.1	72.8	75.8	77.6	78.3	78.1	78.4	79.2	78.3	72.5	63.4
		MCDB	90.1	95.2	92.4	94.0	93.1	91.3	87.5	85.8	88.0	89.0	87.5	90.3
	10%	WB	59.9	63.2	71.3	74.9	77.0	77.8	77.4	77.8	78.4	77.6	70.7	62.1
		MCDB	88.5	94.6	92.5	93.5	92.6	90.6	86.4	84.7	86.9	88.4	87.5	89.0
(35) Mean Daily Temperature Range	5% DB	MDBR	24.1	24.7	23.1	20.1	18.5	16.6	14.3	12.6	15.5	20.9	26.6	25.5
		MCDBR	27.3	27.1	24.7	22.1	20.7	18.8	16.5	14.8	18.1	23.4	27.6	27.6
		MCWBR	10.3	10.1	9.5	8.2	6.0	4.8	5.0	5.8	6.6	6.5	9.0	9.5
	5% WB	MCDBR	26.3	25.5	19.6	18.9	18.1	17.1	15.4	14.0	16.8	19.5	23.4	27.1
		MCWBR	10.8	10.8	9.6	7.0	5.7	5.9	5.9	6.2	7.3	6.9	8.1	9.9
(40) Clear-Sky Solar Irradiance	taub	0.546	0.604	0.725	0.716	0.681	0.643	0.572	0.539	0.536	0.583	0.535	0.508	
	taud	1.837	1.706	1.506	1.611	1.658	1.755	1.949	2.036	2.025	1.882	1.887	1.928	
	Ebn at Noon	240	232	209	210	213	219	236	247	250	235	241	247	
	Edh at Noon	65	77	96	86	80	72	60	55	56	64	61	58	
(44) All-Sky Solar Radiation	RadAvg	1780	1886	1981	1998	1964	1859	1783	1713	1824	1877	1794	1739	
(45)	RadStd	93	95	99	91	59	70	66	68	65	38	54	60	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page